



SYSTEM LEVEL SOLUTIONS (INDIA)

Transformer Monitoring System

For power supply round the clock

About SLS

WE ARE A



Product and Service Company

INTO



Hardware & Software

ESTABLISHED AS



Turnkey Solution provider

Quick Facts

100+

active projects across the globe

300+

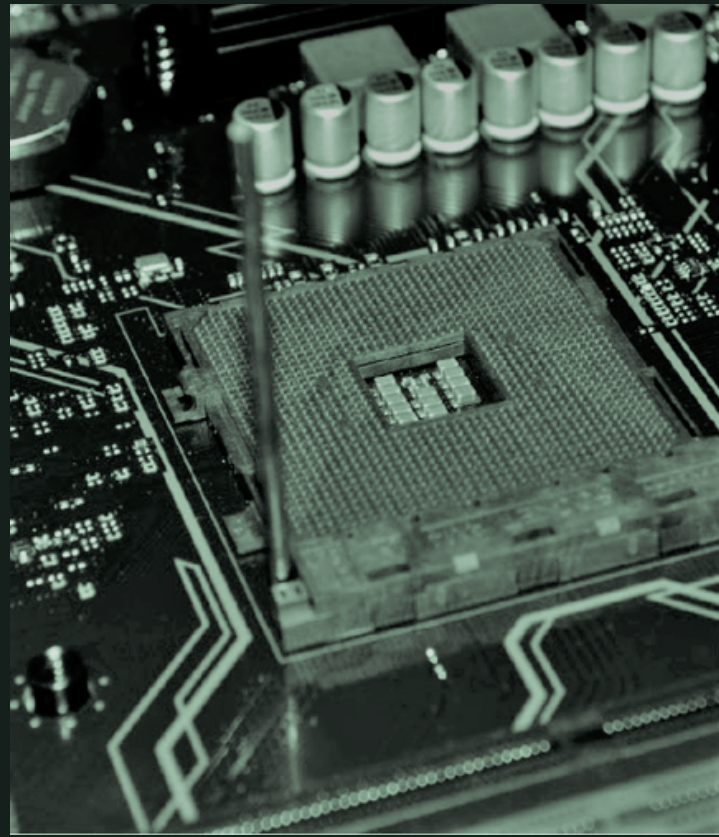
in-house staff

20

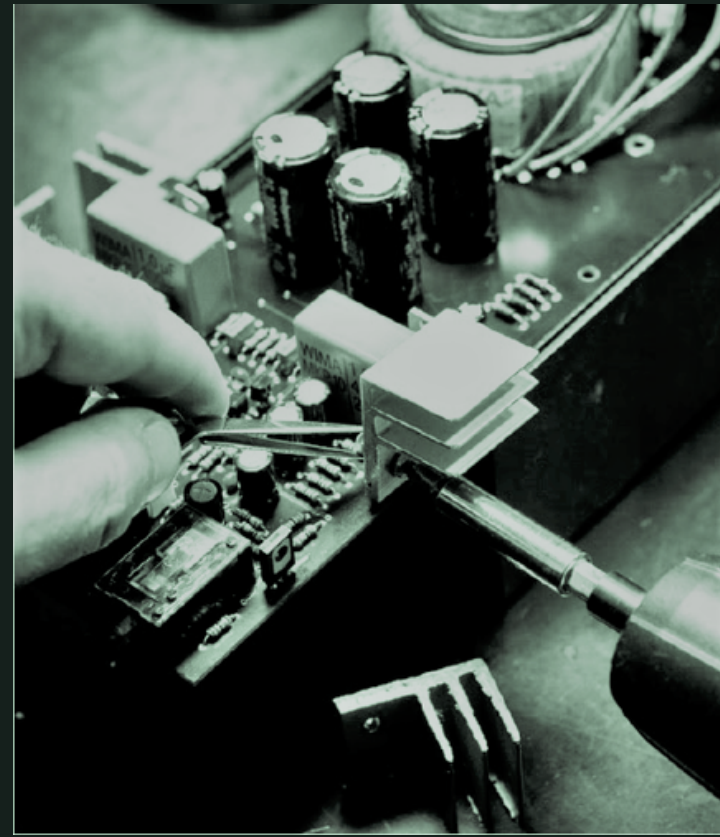
years and counting

- No of business verticals : 6+
- Operating Base: India, UK, USA
- No of manufacturing plants : 1
- In-house Design & Manufacturing Capability
- Part of Make-In-India initiative

Our Core



IP Cores &
FPGA Designs



Hardware
Design



Software
Development

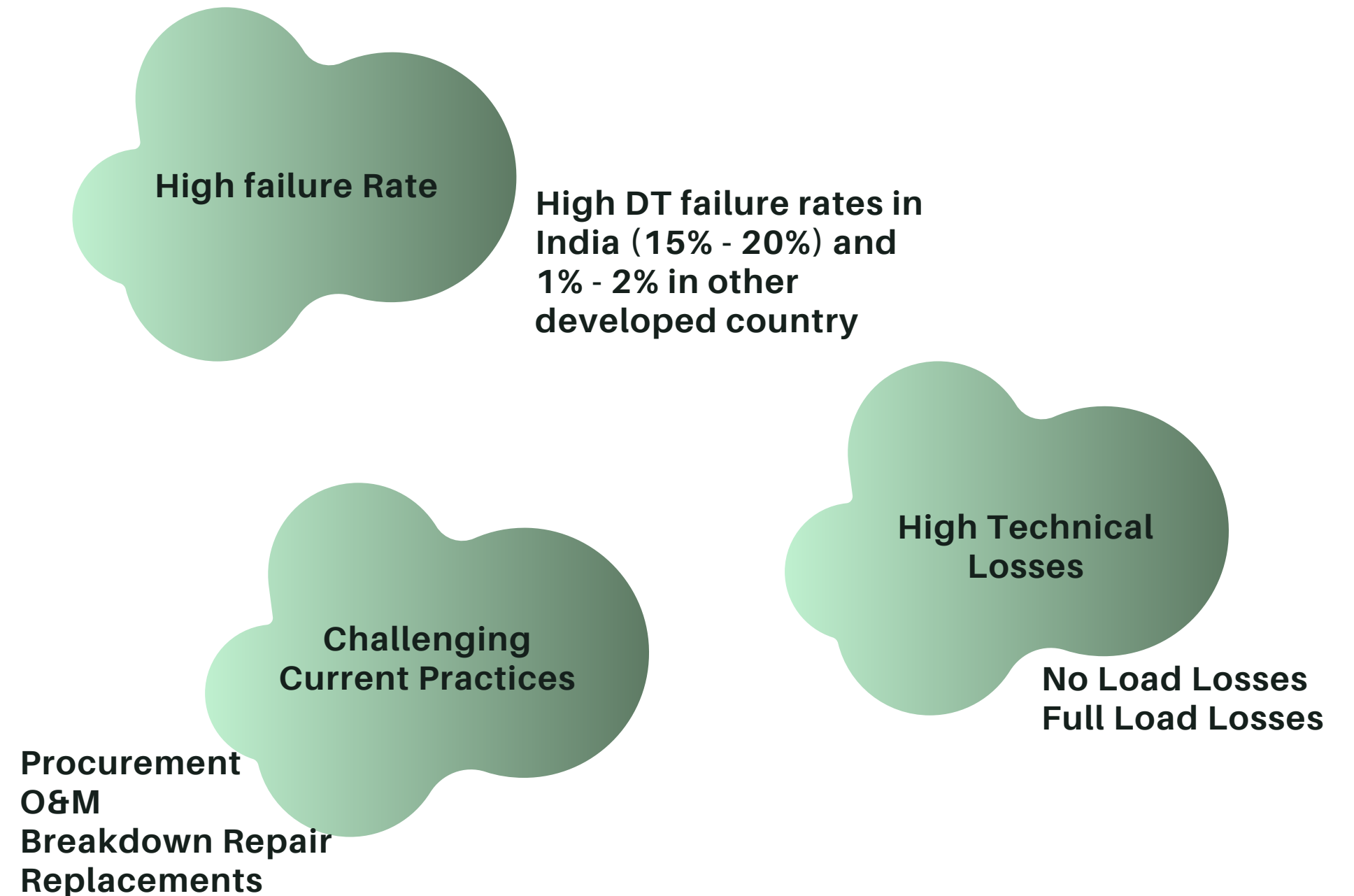


Client
Servicing

**6-8 lakhs DTs fail every year
resulting in huge financial losses
of nearly ₹5000 Cr**

What calls for transformer monitoring?

Issues around DTs





Offerings

TMS CAN MONITOR REAL-TIME ELECTRICAL AND PHYSICAL PARAMETERS

Three phase voltage and current.

Frequency and phase difference.

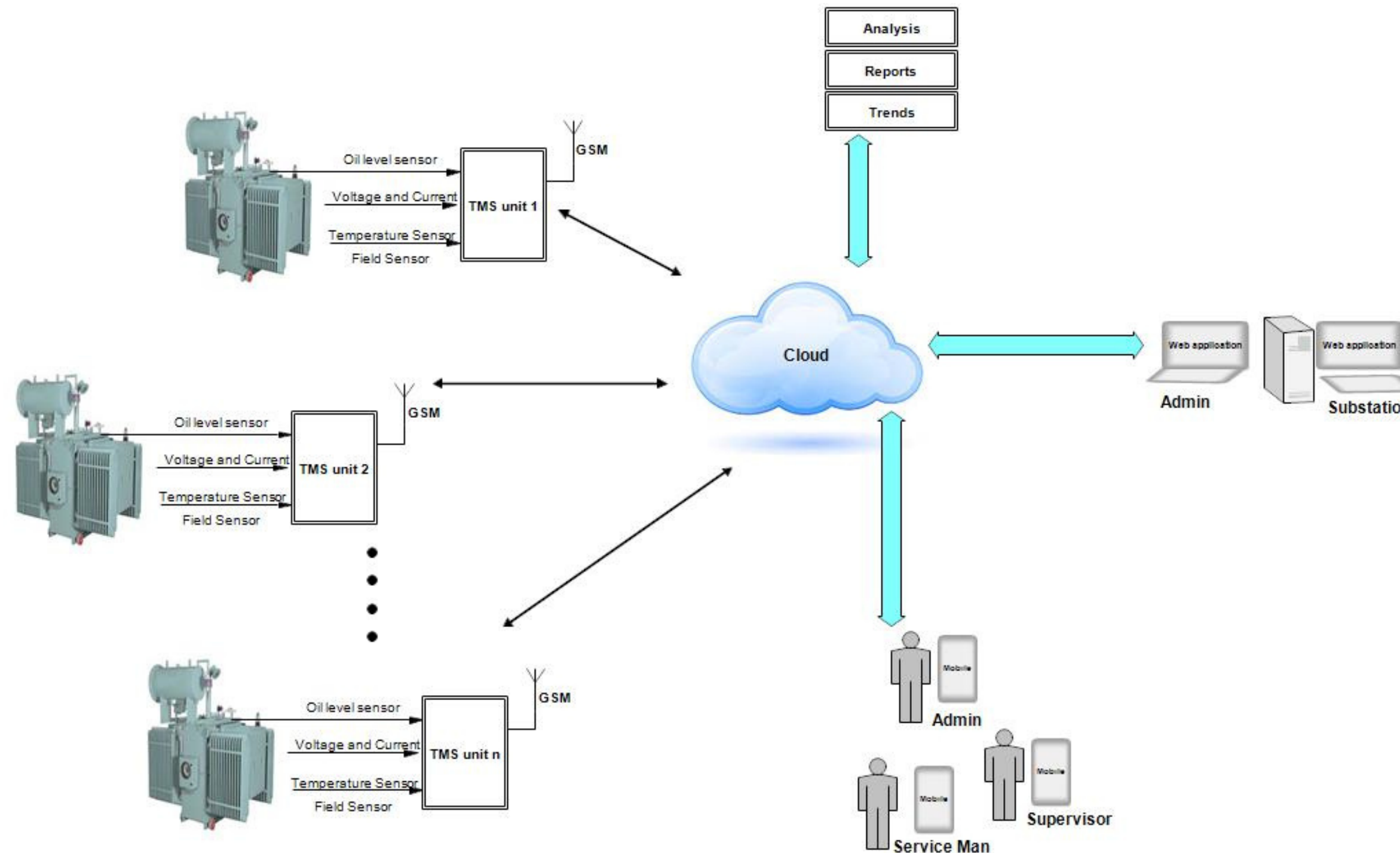
Active Power, Reactive and Apparent power.

Power factor, THD, voltage and current harmonics.

Oil temperature, winding temperature and oil level.

Predict abnormal conditions of transformers.

TMS can send all the energy-related data to a central server for remote monitoring and fault identification purpose





TMS Hardware

UNIT CONSISTS OF A METERING CHIP AND
COMMUNICATION MODULE

IT HAS AN INTERNAL ADC TO SUPPORT
FIELD FUNCTIONALITIES AND SENSORS

SUPPORTS GSM, GPS, BLUETOOTH, UART,
RS485, AND ETHERNET INTERFACES BASED
ON ACTUAL REQUIREMENTS

TWO CHANNELS OF DIGITAL OUTPUT FOR
POTENTIAL FREE CONTACTS



TMS Hardware

FOUR CHANNELS OF OPTICALLY
ISOLATED DIGITAL INPUT FOR 24V

FEW INDICATION LEDS AT THE FRONT
SIDE OF THE UNIT

PROVISION FOR LI-ION BATTERY WITH
BATTERY CHARGING CIRCUIT



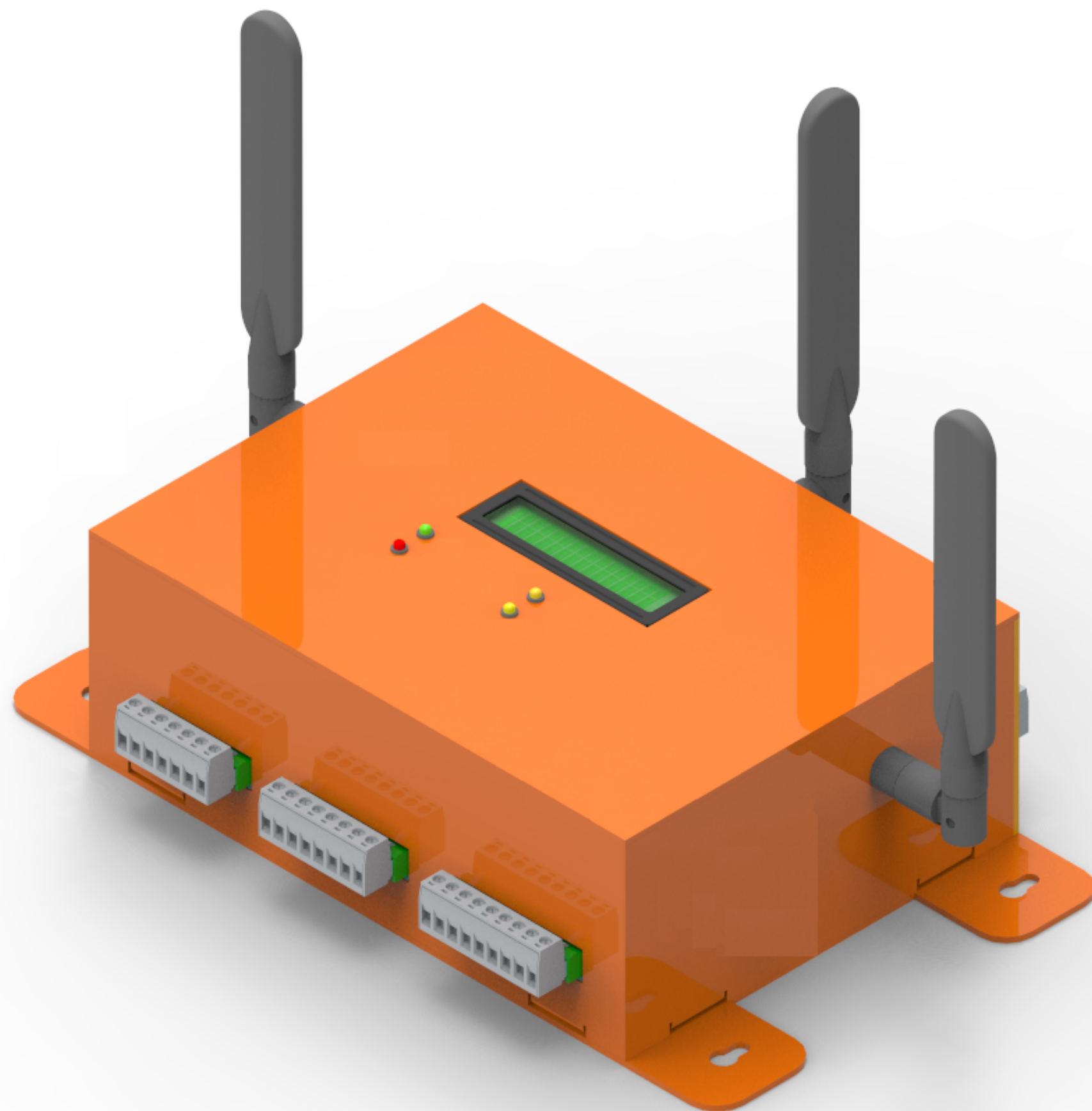
Rogowski coil as a current sensor

Field Sensors



Oil level sensor

PT100 Temperature Sensor



TMS MONITORING BOX

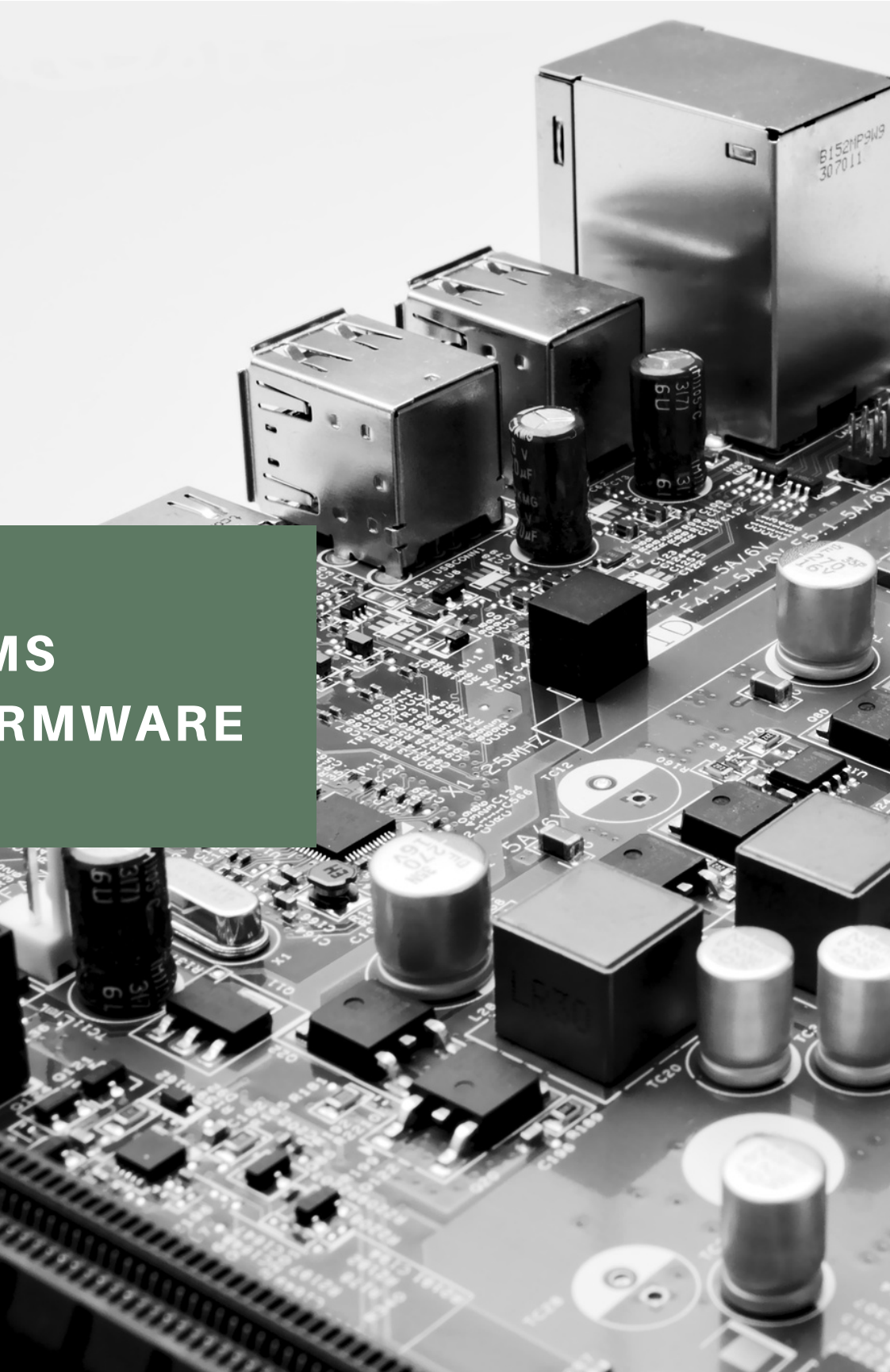


FIRMWARE CAPTURES DATA, VALIDATES
AND PREPARES IT IN SUITABLE FORMAT

TMS SENDS ALL SENSOR AND ENERGY
PARAMETER DATA TO CLOUD ON REGULAR
INTERVALS

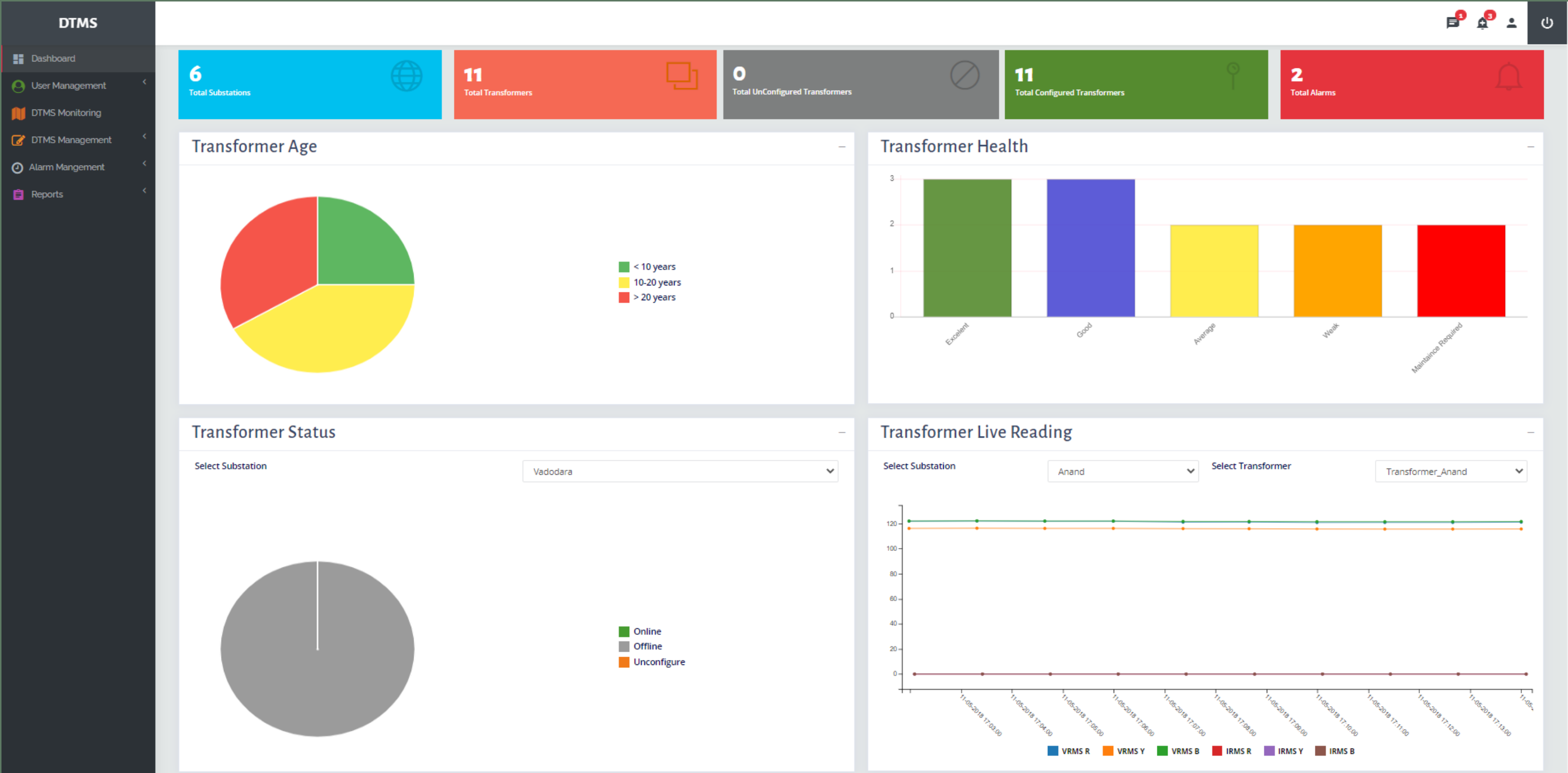
PROVIDES ACCURATE INFORMATION AT
UNIT LEVEL AND PROCESS THIS INFO IN
REAL-TIME

FIRMWARE IS DEVELOPED TO HANDLE
VARIOUS ARITHMETIC AND LOGICAL
OPERATIONS OF ENERGY DATA



**TMS
FIRMWARE**

TMS PORTAL



TMS PORTAL

DTMS

Dashboard

User Management

DTMS Monitoring

DTMS Management

Alarm Mangement

Reports

DTMS Map

+
-

TSM_AHMEDABADz (Offline)

R Phase

Y Phase

B Phase

VRMS: 229.41 V
IRMS: 25 A
P.Angle: 0 deg
Freq.: 49.91 Hz
COSØ: 0.98

VRMS: 229.93V
IRMS: 26.1 A
P.Angle: 120 deg
Freq.: 49.9 Hz
COSØ: 0.98

VRMS: 230.91 V
IRMS: 26.5 A
P.Angle: -120 deg
Freq.: 49.91 Hz
COSØ: 0.98

Oil Temperature in Conservator

Top: 29 degMiddle: 27.5 degBottom: 29 deg

Total Estimated Transformer Life: 25.0 Years

Consumed Transformer Life: 8.0 Years

DTMS Information

Sub Station:

Ahmedabad

Latitude:

22.961583

Longitude:

72.596502

Mac:

00:15:8d:00:01:98:88:ad

Updated On:

27/02/2018 15:21:00

TMS PORTAL



History Report

Select Substation

Select Transformer

Select Date

Select Type

Get Activity

Show 15 entries

PDF Print

Search:

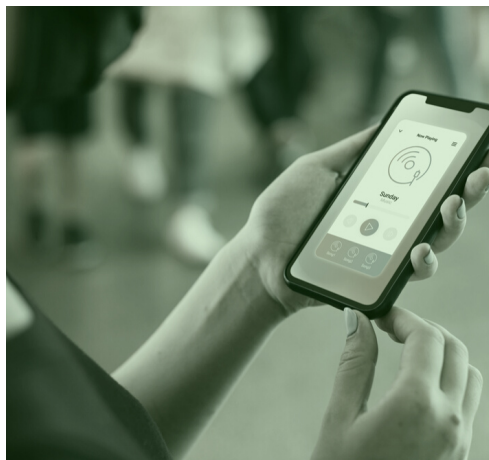
Substation	Transformer	R Phase	Y Phase	B Phase	Communication Time
Anand	Transformer_Anand	121.04	115.63	121.59	2018-11-05 12:44
Anand	Transformer_Anand	120.82	115.54	121.45	2018-11-05 12:42
Anand	Transformer_Anand	120.82	115.54	121.45	2018-11-05 12:41
Anand	Transformer_Anand	120.77	115.61	121.46	2018-11-05 12:39
Anand	Transformer_Anand	121.11	115.8	121.63	2018-11-05 12:38
Anand	Transformer_Anand	121.03	115.87	121.67	2018-11-05 12:37
Anand	Transformer_Anand	121.78	116.07	121.95	2018-11-05 12:35
Anand	Transformer_Anand	121.78	116.07	121.95	2018-11-05 12:34
Anand	Transformer_Anand	121.97	116.22	122.12	2018-11-05 12:33
Anand	Transformer_Anand	121.79	116.08	121.96	2018-11-05 12:31
Anand	Transformer_Anand	121.37	115.81	121.66	2018-11-05 12:30
Anand	Transformer_Anand	120.86	115.65	121.42	2018-11-05 12:29
Anand	Transformer_Anand	120.37	115.17	120.99	2018-11-05 12:27
Anand	Transformer_Anand	120.37	115.17	120.99	2018-11-05 12:26
Anand	Transformer_Anand	120.5	115.29	121.09	2018-11-05 12:25

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REMOTE COMMUNICATION

TMS unit locally via Bluetooth and
central server or cloud via internet connection



EASE OF OPERATIONS

Like Unit installation and Commissioning,
Local data monitoring and Servicing with a
user friendly GUI



IMPORTANT DURING FAULT EMERGENCY

Identify and isolate the faults at actual site
from your phone

MOBILE APPLICATION

The roots of Cost Saving

**FAILURE RATE CAN BE
REDUCE BY 3-5%**



OPERATIONS AND MAINTENANCE SAVING

Detects the failed transformer and
helps to route the O&M



INVENTORY REDUCTION BY 2%

Digitized records of transformers
replaced and to be procured



NO EXTRA DTR METER TO BE INSTALLED

savings of ₹10000- ₹15000 per DT.

THE ROOTS OF COST SAVING

Energy Quality Improvement by monitoring

System Level Solutions

CONSTANT MONITORING OF
TECHNICAL PARAMETERS

Timely maintenance visits and real time
alerts for KPI's

NO-LOAD LOSSES AND FULL
LOAD LOSSES CAN BE
REDUCED

With power quality monitoring



Features

LIVE MONITORING WITH
ALARMS

ENERGY QUALITY
OBSERVATION.

CENTRALISED ACCESS.

INNOVATIVE SOLUTION FOR
SMART GIRDS.



Features

GEOLOCATION BASED
PREDICTIVE ANALYSIS

ENERGY CONSUMPTION
PREDICTION USING AI

INTEGRATION OF ADDITIONAL
SENSORS FOR OIL QUALITY
MONITORING

THANK YOU

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